

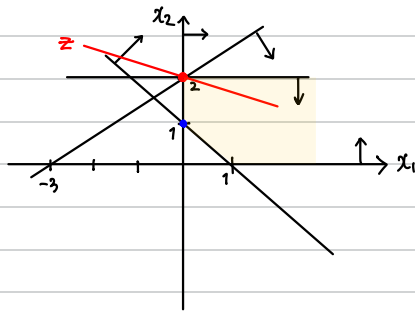
IIE6120 Linear Programming
2025-2 Homework 04
Team B (이정민, 장서현)

[4.1] Consider the following linear programming problem:

$$\begin{array}{ll} \text{Maximize} & -x_1 + 3x_2 \\ \text{subject to} & x_1 + x_2 \geq 1 \\ & -2x_1 + 3x_2 \leq 6 \\ & x_2 \leq 2 \\ & x_1, x_2 \geq 0. \end{array}$$

- a. Solve the problem graphically.
- b. Solve the problem by the two-phase simplex method. Show that the points generated by Phase I correspond to basic solutions of the original system.

a.



$$(x_1^*, x_2^*) = (0, 2) \\ z^* = 6$$

b.

$$\text{Max } z = -x_1 + 3x_2$$

$$\begin{array}{rcll} \text{s.t.} & x_1 + x_2 - x_3 & + x_6 & = 1 \\ & -2x_1 + 3x_2 & + x_4 & = 6 \\ & x_2 & + x_5 & = 2 \\ & x_1, x_2, x_3, x_4, x_5, x_6 & \geq 0 \end{array}$$

Phase 1. $\text{Min } z = x_6$

□	z	x_1	x_2	x_3	x_4	x_5	x_6	RHS
z	1	0	0	0	0	0	-1	0
x_6	0	1	1	-1	0	0	1	1
x_4	0	-2	3	0	1	0	0	6
x_5	0	0	1	0	0	1	0	2

□	z	x_1	x_2	x_3	x_4	x_5	x_6	RHS
z	1	<u>1</u>	<u>1</u>	-1	0	0	0	1
x_6	0	1	<u>1</u>	-1	0	0	1	<u>1</u>
x_4	0	-2	3	0	1	0	0	6
x_5	0	0	1	0	0	1	0	2

□	z	x_1	x_2	x_3	x_4	x_5	x_6	RHS
z	1	0	0	0	0	0	-1	0
x_2	0	1	1	-1	0	0	1	1
x_4	0	-5	0	3	1	0	-3	3
x_5	0	-1	0	1	0	1	-1	1

$$x_B = \{x_2, x_4, x_5\}$$

$$(x_1, x_2) = (0, 1)$$

Phase 2. $\text{Max } z = -x_1 + 3x_2$

□	z	x_1	x_2	x_3	x_4	x_5	RHS
z	1	4	0	-3	0	0	3
x_2	0	1	1	-1	0	0	1
x_4	0	-5	0	<u>3</u>	1	0	3
x_5	0	-1	0	1	0	1	1

□	z	x_1	x_2	x_3	x_4	x_5	RHS
z	1	<u>-1</u>	0	0	1	0	6
x_2	0	-2/3	1	0	1/3	0	2
x_3	0	-5/3	0	1	1/3	0	1
x_5	0	2/3	0	0	-1/3	1	0

□	z	x_1	x_2	x_3	x_4	x_5	RHS
z	1	0	0	0	1/2	3/2	6
x_2	0	0	1	0	0	1	2
x_3	0	0	0	1	-1/2	5/2	1
x_1	0	1	0	0	-1/2	3/2	0

$$x_B^* = \{x_1, x_2, x_3\}$$

$$(x_1^*, x_2^*) = (0, 2)$$

$$z^* = 6$$

[4.2] Solve the following problem by the two-phase simplex method:

$$\begin{aligned} \text{Minimize } & x_1 + 3x_2 - x_3 \\ \text{subject to } & 2x_1 + x_2 + 3x_3 \geq 3 \\ & -x_1 + x_2 \geq 1 \\ & -x_1 - 5x_2 + x_3 \leq 4 \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

$$\text{Min } z = x_1 + 3x_2 - x_3$$

$$2x_1 + x_2 + 3x_3 - x_4 + x_7 = 3$$

$$-x_1 + x_2 - x_5 + x_8 = 1$$

$$-x_1 - 5x_2 + x_3 + x_6 = 4$$

$$x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8 \geq 0$$

Phase 1. Min $z = x_7 + x_8$

	z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
z	1	0	0	0	0	0	0	-1	-1	0
x_7	0	2	1	3	-1	0	0	1	0	3
x_8	0	-1	1	0	0	-1	0	0	1	1
x_6	0	-1	-5	1	0	0	1	0	0	4

	z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
z	1	1	2	3	-1	-1	0	0	0	4
x_7	0	2	1	3	-1	0	0	1	0	3
x_8	0	-1	1	0	0	-1	0	0	1	1
x_6	0	-1	-5	1	0	0	1	0	0	4

	z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
z	1	-1	1	0	0	-1	0	-1	0	1
x_3	0	2/3	1/3	1	-1/3	0	0	1/3	0	1
x_8	0	-1	1	0	0	-1	0	0	1	1
x_6	0	-5/3	-16/3	0	1/3	0	1	-1/3	0	3

	z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
z	1	0	0	0	0	0	0	-1	-1	0
x_3	0	1	0	1	-1/3	1/3	0	1/3	-1/3	2/3
x_2	0	-1	1	0	0	-1	0	0	1	1
x_6	0	-7	0	0	1/3	-16/3	1	-1/3	16/3	25/3

$$X_{BV} = \{x_3, x_2, x_6\}$$

Phase 2. Min $z = x_1 + 3x_2 - x_3$

	z	x_1	x_2	x_3	x_4	x_5	x_6	RHS
z	1	-5	0	0	1/3	-10/3	0	2/3
x_3	0	1	0	1	-1/3	1/3	0	2/3
x_2	0	-1	1	0	0	-1	0	1
x_6	0	-7	0	0	1/3	-16/3	1	25/3

	z	x_1	x_2	x_3	x_4	x_5	x_6	RHS
z	1	12	0	0	0	2	-1	-6
x_3	0	-6	0	1	0	-5	1	9
x_2	0	-1	1	0	0	-1	0	1
x_4	0	-21	0	0	1	-16	3	25

↓
this model is unbounded.

[4.10] Use the big-M method to solve the following problem:

$$\begin{aligned} \text{Minimize} \quad & -2x_1 + 2x_2 + x_3 + x_4 \\ \text{subject to} \quad & x_1 + 2x_2 + 2x_3 + x_4 \leq 2 \\ & x_1 - 2x_2 + x_3 + 2x_4 \geq 3 \\ & 2x_1 - x_2 + 3x_3 \geq 2 \\ & x_1, x_2, x_3, x_4 \geq 0. \end{aligned}$$

(M is a big positive number)

Q3. (4.10)

$$\begin{aligned} \text{Minimize} \quad & -2x_1 + 2x_2 + x_3 + x_4 + Mx_8 + Mx_9 \\ \text{s.t.} \quad & x_1 + 2x_2 + 2x_3 + x_4 + x_5 = 2 \\ & x_1 - 2x_2 + x_3 + 2x_4 - x_6 + x_8 = 3 \\ & 2x_1 - x_2 + 3x_3 - x_7 + x_9 = 2 \\ & x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8, x_9 \geq 0 \end{aligned}$$

	z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	RHS
z	1	2	-2	-1	-1	0	0	0	-M	-M	0
x_5	0	1	2	2	1	1	0	0	0	0	2
$\bar{x}_8$	0	1	-2	1	2	0	-1	0	1	0	3
$\bar{x}_9$	0	2	-1	3	0	0	0	-1	0	1	2
z	1	$2+3M$	$-2-3M$	$-1+4M$	$-1+2M$	0	-M	-M	0	0	$5M$
x_5	0	1	2	2	1	1	0	0	0	0	2
$\bar{x}_8$	0	1	-2	1	2	0	-1	0	1	0	3
$\bar{x}_9$	0	2	-1	3	0	0	0	-1	0	1	2
z	1	$\frac{8}{3} + \frac{1}{3}M$	$-\frac{10}{3} - \frac{5}{3}M$	0	$-1+2M$	0	-M	$\frac{1}{3} - \frac{1}{3}M$	0	$\frac{1}{3} - \frac{1}{3}M$	$\frac{2}{3} + \frac{1}{3}M$
x_5	0	$-1/3$	$8/3$	0	1	1	0	$2/3$	0	$-2/3$	$2/3$
$\bar{x}_8$	0	$1/3$	$-5/3$	0	2	0	-1	$1/3$	1	0	$11/3$
x_3	0	$2/3$	$-1/3$	1	0	0	0	$-1/3$	0	$1/3$	$2/3$
z	1	$\frac{11}{3} + M$	$\frac{1}{3} - 11M$	0	0	$1+2M$	-M	$\frac{1}{3} - M$	0	$-1/3$	$\frac{4}{3} + M$
x_4	0	$-1/3$	$8/3$	0	1	1	0	$2/3$	0	$-2/3$	$2/3$
$\bar{x}_8$	0	1	-11	0	0	-2	-1	-1	1	1	1
x_3	0	$2/3$	$-1/3$	1	0	0	0	$-1/3$	0	$1/3$	$2/3$
z	1	0	$50/3$	0	0	$11/3$	$11/3$	$8/3$	$-M - \frac{1}{3}$	$-M - \frac{8}{3}$	-1
x_4	0	0	$1/3$	0	1	$1/3$	$-1/3$	$1/3$	$1/3$	$-1/3$	1
x_1	0	1	-11	0	0	-2	-1	-1	1	1	1
x_3	0	0	$13/3$	1	0	$4/3$	$2/3$	$1/3$	$-2/3$	$-1/3$	0
z	1	0	0	$-50/3$	0	$11/3$	$-3/3$	$18/3$	$\frac{2}{3} - M$	$-M - \frac{18}{3}$	-1
x_4	0	0	0	$-1/3$	1	$3/3$	$-5/3$	$4/3$	$5/3$	$-4/3$	1
x_1	0	1	0	$21/3$	0	$2/3$	$1/3$	$-6/3$	$-1/3$	$6/3$	1
x_2	0	0	1	$31/3$	0	$4/3$	$2/3$	$1/3$	$-2/3$	$-1/3$	0
z	1	0	-18	-8	0	-5	-3	0	$3-M$	$-M$	-1
x_4	0	0	-4	-1	1	-1	-1	0	1	0	1
x_1	0	1	6	3	0	2	1	0	-1	0	1
x_2	0	0	13	3	0	4	2	1	-2	-1	0

$$x_1 = 1, x_4 = 1, z^* = -1$$

[4.35] Solve the following problem, using the lexicographic rule for noncycling.

Repeat using Bland's Rule:

#4

$$\begin{array}{llll} \text{Maximize} & x_1 & + & 2x_2 & + & x_3 \\ \text{subject to} & x_1 & + & 4x_2 & + & 3x_3 \leq 4 \\ & -x_1 & + & x_2 & + & 4x_3 \leq 1 \\ & x_1 & + & 3x_2 & + & x_3 \leq 6 \\ & x_1, & & x_2, & & x_3 \geq 0. \end{array}$$

$$\begin{array}{ll} \text{Max } z = & x_1 + 2x_2 + x_3 \\ \text{s.t.} & x_1 + 4x_2 + 3x_3 + s_1 = 4 \\ & -x_1 + x_2 + 4x_3 + s_2 = 1 \\ & x_1 + 3x_2 + x_3 + s_3 = 6 \\ & x_1, x_2, x_3, s_1, s_2, s_3 \geq 0 \end{array}$$

[Lexicographic Rule]

1	z	x_1	x_2	x_3	s_1	s_2	s_3	RHS	b/y	q/y
z	1	-1	-2	-1	0	0	0	0		
s_1	0	1	4	3	1	0	0	4	1	1/4
s_2	0	-1	1	4	0	1	0	1	1	-1
s_3	0	1	3	1	0	0	1	6	2	

2	z	x_1	x_2	x_3	s_1	s_2	s_3	RHS	b/y
z	1	-3	0	7	0	2	0	2	
s_1	0	5	0	-13	1	-4	0	0	0
x_2	0	-1	1	4	0	1	0	1	
s_3	0	4	0	-11	0	-3	1	3	3/4

3	z	x_1	x_2	x_3	s_1	s_2	s_3	RHS	b/y
z	1	0	0	-4/5	3/5	-2/5	0	2	
x_1	0	1	0	-13/5	1/5	-4/5	0	0	
x_2	0	0	1	1/5	1/5	1/5	0	1	5/7
s_3	0	0	0	-3/5	-4/5	1/5	1	3	

4	z	x_1	x_2	x_3	s_1	s_2	s_3	RHS	b/y
z	1	0	4/7	0	5/7	-2/7	0	18/7	
x_1	0	1	13/7	0	4/7	-3/7	0	13/7	
x_3	0	0	5/7	1	1/7	1/7	0	5/7	5
s_3	0	0	3/7	0	-5/7	2/7	1	24/7	12

5	z	x_1	x_2	x_3	s_1	s_2	s_3	RHS	→ optimal $x_{BV} = \{x_1, s_2, s_3\}$	
z	1	0	2	2	1	0	0	4	$x_1 = 4, s_2 = 5, s_3 = 2, z^* = 4$	
x_1	0	1	4	3	1	0	0	4		
s_2	0	0	5	7	1	1	0	5		
s_3	0	0	-1	-2	-1	0	1	2		

[Bland's Rule]

1	z	x_1	x_2	x_3	s_1	s_2	s_3	RHS	→	2	z	x_1	x_2	x_3	s_1	s_2	s_3	RHS
z	1	-1	-2	-1	0	0	0	0		z	1	0	2	2	1	0	0	4
s_1	0	1	4	3	1	0	0	4		x_1	0	1	4	3	1	0	0	4
s_2	0	-1	1	4	0	1	0	1		s_2	0	0	5	7	1	1	0	5
s_3	0	1	3	1	0	0	1	6		s_3	0	0	-1	-2	-1	0	1	2

→ optimal $x_{BV} = \{x_1, s_2, s_3\}$
 $x_1 = 4, s_2 = 5, s_3 = 2, z^* = 4$

[5.1] Solve the following linear program by the revised simplex method:

#5 Minimize $x_1 + 4x_2 - 7x_3 + x_4 + 5x_5$
 subject to $x_1 - (3/4)x_2 + 2x_3 - (1/4)x_4 = 6$
 $-(1/4)x_2 + 3x_3 - (3/4)x_4 + x_5 = 5$
 $x_1, x_2, x_3, x_4, x_5 \geq 0.$

Q5. (5.1)

Minimize $x_1 + 4x_2 - 7x_3 + x_4 + 5x_5 + M\bar{x}_6 + M\bar{x}_7$
 s.t $x_1 - \frac{3}{4}x_2 + 2x_3 - \frac{1}{4}x_4 + \bar{x}_6 = 6$
 $-\frac{1}{4}x_2 + 3x_3 - \frac{3}{4}x_4 + \bar{x}_7 = 5$
 $x_1, x_2, x_3, x_4, x_5, \bar{x}_6, \bar{x}_7 \geq 0.$

Initial

$B = [\bar{x}_6, \bar{x}_7] = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = B^{-1}$
 $w = C_B B^{-1} = [M, M] \quad \bar{b} = B^{-1}b = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 6 \\ 5 \end{bmatrix} = \begin{bmatrix} 6 \\ 5 \end{bmatrix}$

	B^{-1}	RHS
Z	0 0	-11M $\rightarrow (-1M)x_1 + (-4M)x_2 + (-5M)x_3$
$\bar{x}_6$	1 0	6 $+ (1+M)x_4 + (-5-M)x_5$
$\bar{x}_7$	0 1	5

Iteration 1

$Z_1 - C_1 = [M, M] \begin{bmatrix} 1 \\ 0 \end{bmatrix} - 1 = M - 1$
 $Z_2 - C_2 = \begin{bmatrix} 3/4 \\ 1/4 \end{bmatrix} - 4 = -M - 4$
 $Z_3 - C_3 = \begin{bmatrix} 2 \\ 3 \end{bmatrix} + M = 5M + 7 \rightarrow x_3: e.v$
 $Z_4 - C_4 = \begin{bmatrix} -1/4 \\ -3/4 \end{bmatrix} - 1 = -M - 1$
 $Z_5 - C_5 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - 5 = -5$
 $y_3 = B^{-1}a_3 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$

	B^{-1}	RHS	x_3
Z	0 0	-11M	-11M
$\bar{x}_6$	1 0	6	2 $1/3$
$\bar{x}_7$	0 1	5	3 $3/5 \rightarrow$ leaving var

Iteration 2

$B = [\bar{x}_6, x_3] = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \quad B^{-1} = \begin{bmatrix} 1 & -2/3 \\ 0 & 1/3 \end{bmatrix}$
 $w = C_B B^{-1} = [M, -11] \begin{bmatrix} 1 & -2/3 \\ 0 & 1/3 \end{bmatrix}$
 $= [M, -\frac{2}{3}M - \frac{11}{3}]$
 $\bar{b} = B^{-1}\bar{b} = \begin{bmatrix} 1 & -2/3 \\ 0 & 1/3 \end{bmatrix} \begin{bmatrix} 6 \\ 5 \end{bmatrix} = \begin{bmatrix} 6 - \frac{10}{3} \\ \frac{5}{3} \end{bmatrix} = \begin{bmatrix} 8/3 \\ 5/3 \end{bmatrix}$

	B^{-1}	RHS
Z	0 0	$-\frac{11}{3} - \frac{2}{3}M$
$\bar{x}_6$	1 -2/3	8/3
x_3	0 1/3	5/3

$-11M + 5 \times (-\frac{2}{3} - \frac{2}{3}M) = \frac{25}{3} - \frac{58}{3}M$

$Z_1 - C_1 = [M, -\frac{2}{3}M - \frac{11}{3}] \begin{bmatrix} 1 \\ 0 \end{bmatrix} - 1 = M - 1 \rightarrow e.v$
 $Z_2 - C_2 = \begin{bmatrix} 3/4 \\ 1/4 \end{bmatrix} - 4 = -\frac{3}{4}M + \frac{2}{3}M + \frac{11}{3} - 4 = -\frac{1}{12}M - \frac{1}{12}$
 $Z_4 - C_4 = \begin{bmatrix} -1/4 \\ -3/4 \end{bmatrix} - 1 = -\frac{1}{4}M + \frac{1}{2}M + \frac{11}{4} - 1 = \frac{1}{4}M + \frac{7}{4}$
 $Z_5 - C_5 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - 5 = -5$
 $Z_7 - C_7 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - M = -\frac{2}{3}M - \frac{11}{3} - M = -\frac{5}{3}M - \frac{11}{3}$
 $y_1 = B^{-1}a_1 = \begin{bmatrix} 1 & -2/3 \\ 0 & 1/3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$
 $\bar{x}_6$ 8/3 1 $2/8 \rightarrow i.v$
 x_3 5/3 0 0

Iteration 3

$B = [x_1, x_3] = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \quad B^{-1} = \begin{bmatrix} 1 & -2/3 \\ 0 & 1/3 \end{bmatrix}$
 $w = C_B B^{-1} = [1, -11] \begin{bmatrix} 1 & -2/3 \\ 0 & 1/3 \end{bmatrix} = [1, -\frac{2}{3} - \frac{11}{3}] = [1, -13]$
 $\bar{b} = B^{-1}\bar{b} = \begin{bmatrix} 1 & -2/3 \\ 0 & 1/3 \end{bmatrix} \begin{bmatrix} 6 \\ 5 \end{bmatrix} = \begin{bmatrix} 6 - \frac{10}{3} \\ \frac{5}{3} \end{bmatrix} = \begin{bmatrix} 8/3 \\ 5/3 \end{bmatrix}$

B^{-1} RHS

	B^{-1}	RHS
Z	1 -3	-9
x_1	1 -2/3	8/3
x_3	0 1/3	5/3

$Z_2 - C_2 = \begin{bmatrix} 3/4 \\ 1/4 \end{bmatrix} - 4 = -\frac{3}{4} + \frac{3}{4} = 0$
 $Z_4 - C_4 = \begin{bmatrix} -1/4 \\ -3/4 \end{bmatrix} - 1 = -\frac{1}{4} + \frac{9}{4} = 2 \rightarrow e.v$
 $Z_5 - C_5 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - 5 = -5 < 0$
 $Z_6 - C_6 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} - M = 1 - M$
 $Z_7 - C_7 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} - M = -M$
 $y_4 = B^{-1}a_4 = \begin{bmatrix} 1 & -2/3 \\ 0 & 1/3 \end{bmatrix} \begin{bmatrix} -1/4 \\ -3/4 \end{bmatrix} = \begin{bmatrix} -\frac{1}{4} + \frac{1}{2} \\ -\frac{1}{4} \end{bmatrix} = \begin{bmatrix} 1/4 \\ -1/4 \end{bmatrix}$
 x_1 8/3 1/2 $\rightarrow e.v$
 x_3 5/3 -1/4

Iteration 4

$B = [x_4, x_3] = \begin{bmatrix} -1/4 & 2 \\ -3/4 & 3 \end{bmatrix} \quad B^{-1} = \begin{bmatrix} 4 & -8/3 \\ 1 & -1/3 \end{bmatrix}$
 $w = C_B B^{-1} = [1, -11] \begin{bmatrix} 4 & -8/3 \\ 1 & -1/3 \end{bmatrix} = \begin{bmatrix} 4 - 11 \\ -\frac{8}{3} + \frac{11}{3} \end{bmatrix} = \begin{bmatrix} -7 \\ \frac{3}{3} \end{bmatrix} = \begin{bmatrix} -7 \\ 1 \end{bmatrix}$
 $Z_1 - C_1 = \begin{bmatrix} -7 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} - 1 = -7 - 1 < 0$
 $Z_2 - C_2 = \begin{bmatrix} 3/4 \\ 1/4 \end{bmatrix} - 4 = \frac{9}{4} + \frac{1}{4} - 4 < 0$
 $Z_5 - C_5 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - 5 < 0$
 $Z_6 - C_6 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} - M < 0$
 $Z_7 - C_7 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} - M < 0$
 $Z_3 - C_3 \leq 0 \quad \forall \text{ n.b.v} \Rightarrow \text{STOP}$
 $\bar{b} = B^{-1}\bar{b} = \begin{bmatrix} 4 & -8/3 \\ 1 & -1/3 \end{bmatrix} \begin{bmatrix} 6 \\ 5 \end{bmatrix} = \begin{bmatrix} 24 - \frac{40}{3} \\ 6 - \frac{5}{3} \end{bmatrix} = \begin{bmatrix} \frac{12-40}{3} \\ \frac{18-5}{3} \end{bmatrix} = \begin{bmatrix} -\frac{28}{3} \\ \frac{13}{3} \end{bmatrix}$
 $\therefore (x_1, x_2, x_3, x_4, x_5) = (0, 0, \frac{13}{3}, \frac{28}{3}, 0)$
 $z = -\frac{59}{3}$

[5.8] Apply the revised simplex method to the following problem, both with and without the lexicographic cycling prevention rule. (This is the example used in Chapter 4 to illustrate cycling.)

#6

$$\begin{array}{llllll} \text{Minimize} & & - (3/4)x_4 & + 20x_5 & - (1/2)x_6 & + 6x_7 \\ \text{subject to} & x_1 & + (1/4)x_4 & - 8x_5 & - x_6 & + 9x_7 = 0 \\ & x_2 & + (1/2)x_4 & - 12x_5 & - (1/2)x_6 & + 3x_7 = 0 \\ & & & & + x_6 & = 1 \\ & x_1, x_2, x_3, & x_4, & x_5, & x_6, & x_7 \geq 0. \end{array}$$

Without lexicographic cycling (Following Ex 4.11)

①		B^{-1}	RHS	
z	0	0	0	0
x_1	1	0	0	0
x_2	0	1	0	0
x_3	0	0	1	1

$$z_4 - c_4 = 3/4 \rightarrow y_4 = \begin{bmatrix} 1/4 \\ 1/2 \\ 0 \end{bmatrix}$$

$$z_5 - c_5 = -20$$

$$z_6 - c_6 = 1/2$$

$$z_7 - c_7 = -6$$

②		B^{-1}	RHS	
z	-3	0	0	0
x_4	4	0	0	0
x_2	-2	1	0	0
x_3	0	0	1	1

$$z_1 - c_1 = -3 \rightarrow y_5 = \begin{bmatrix} -3/2 \\ 4 \\ 0 \end{bmatrix}$$

$$z_5 - c_5 = 4$$

$$z_6 - c_6 = 7/2$$

$$z_7 - c_7 = -33$$

③		B^{-1}	RHS	
z	-1	-1	0	0
x_4	-12	8	0	0
x_5	-1/2	1/4	0	0
x_3	0	0	1	1

$$z_1 - c_1 = -1 \rightarrow y_6 = \begin{bmatrix} 8 \\ 3/8 \\ 1 \end{bmatrix}$$

$$z_2 - c_2 = -1$$

$$z_6 - c_6 = 2$$

$$z_7 - c_7 = -18$$

④		B^{-1}	RHS	
z	2	-3	0	0
x_6	-3/2	1	0	0
x_5	1/16	-1/8	0	0
x_3	3/2	-1	1	1

$$z_1 - c_1 = 2 \rightarrow y_7 = \begin{bmatrix} -2/2 \\ 3/16 \\ 2/2 \end{bmatrix}$$

$$z_2 - c_2 = -3$$

$$z_4 - c_4 = -1/4$$

$$z_7 - c_7 = 3$$

⑤		B^{-1}	RHS	
z	1	-1	0	0
x_6	2	-6	0	0
x_7	1/3	-2/3	0	0
x_3	-2	6	1	1

$$z_1 - c_1 = 1 \rightarrow y_1 = \begin{bmatrix} 2 \\ 1/3 \\ -2 \end{bmatrix}$$

$$z_2 - c_2 = -1$$

$$z_4 - c_4 = 0$$

$$z_5 - c_5 = 1/2$$

⑥		B^{-1}	RHS	
z	0	2	0	0
x_1	1	-3	0	0
x_7	0	1/3	0	0
x_3	0	0	1	1

$$z_2 - c_2 = 2 \rightarrow y_2 = \begin{bmatrix} -3 \\ 1/3 \\ 0 \end{bmatrix}$$

$$z_4 - c_4 = 7/4$$

$$z_5 - c_5 = -44$$

$$z_6 - c_6 = -1/2$$

⑦		B^{-1}	RHS	
z	0	0	0	0
x_1	1	0	0	0
x_2	0	1	0	0
x_3	0	0	1	1

$$z_4 - c_4 = 3/4 \rightarrow y_4 = \begin{bmatrix} 1/4 \\ 1/2 \\ 0 \end{bmatrix}$$

$$z_5 - c_5 = -20$$

$$z_6 - c_6 = 1/2$$

$$z_7 - c_7 = -6$$

cycling!
Using the lexicographic
cycling rule,



⑤		B^{-1}	RHS
z	0	$-\frac{3}{2}$ 0	0
x_1	1	$-\frac{1}{2}$ 0	0
x_4	0	2 0	0
x_3	0	0 1	1

$$z_2 - C_2 = (0, -\frac{3}{2}, 0) \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = -\frac{3}{2}$$

$$z_5 - C_5 = (0, -\frac{3}{2}, 0) \begin{pmatrix} 0 \\ -\frac{1}{2} \\ 0 \end{pmatrix} - 20 = -2$$

$$z_6 - C_6 = (0, -\frac{3}{2}, 0) \begin{pmatrix} 0 \\ -\frac{1}{2} \\ 1 \end{pmatrix} - (-\frac{1}{2}) = \frac{5}{4}$$

$$z_7 - C_7 = (0, -\frac{3}{2}, 0) \begin{pmatrix} 0 \\ \frac{1}{2} \\ 0 \end{pmatrix} - 6 = -\frac{11}{2}$$

$$\rightarrow y_6 = \begin{bmatrix} 1 & -1/2 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ -1/2 \\ 1 \end{bmatrix} = \begin{bmatrix} -3/4 \\ -1 \\ 1 \end{bmatrix}$$

⑥		B^{-1}	RHS
z	0	$-3/2$ $-5/4$	$-5/4$
x_1	1	$-1/2$ $3/4$	$3/4$
x_4	0	2 1	0
x_6	0	0 1	1

$$z_2 - C_2 = (0, -\frac{3}{2}, -\frac{5}{4}) \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = -\frac{3}{2}$$

$$z_3 - C_3 = (0, -\frac{3}{2}, -\frac{5}{4}) \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = -\frac{5}{4}$$

$$z_5 - C_5 = (0, -\frac{3}{2}, -\frac{5}{4}) \begin{pmatrix} 0 \\ -\frac{1}{2} \\ 0 \end{pmatrix} - 20 = -2$$

$$z_7 - C_7 = (0, -\frac{3}{2}, -\frac{5}{4}) \begin{pmatrix} 0 \\ \frac{1}{2} \\ 0 \end{pmatrix} - 6 = -\frac{11}{2}$$

$\rightarrow$ optimal!
 $z^* = -5/4$
 $x_1 = 3/4, x_4 = 0, x_6 = 1$
 $x_2 = x_3 = x_5 = 0$

[5.14] Solve the following problem by the simplex method for bounded variables:

#7

$$\begin{aligned} \text{Maximize } & 2x_1 + 3x_2 - 2x_3 \\ \text{subject to } & x_1 + 3x_2 + x_3 \leq 8 \\ & 2x_1 + x_2 - x_3 \geq 3 \\ & x_1 \leq 4 \\ & -2 \leq x_2 \leq 3 \\ & x_3 \geq 2. \end{aligned}$$

Optimal objective 6.000000000e+00

Optimal Solution:

$x_1 = 4.0000$

$x_2 = 0.6667$

$x_3 = 2.0000$

Objective Value = 6.0000

$$\begin{aligned} \text{Minimize } & -2d_1 - 3d_2 + 2d_3 \\ \text{s.t. } & d_1 + 3d_2 + d_3 + s = 8 \\ & 2d_1 + d_2 - d_3 - e + a = 3 \\ & 0 \leq d_1 \leq 4 \\ & -2 \leq d_2 \leq 3 \\ & 2 \leq d_3 \leq \infty \\ & s, e, a \geq 0. \end{aligned}$$

Initial BFS

$$d_1(l)=0, d_2(l)=-2, d_3(l)=2, e(l)=0$$

$$0 - 6 + 2 + s = 8 \Rightarrow s = 12$$

$$0 - 2 - 2 - 0 + a = 3 \Rightarrow a = 7$$

Phase 1. Minimize $z = a$

$$\begin{array}{c|ccccccc|c} z & d_1^l & d_2^l & d_3^l & s & e^l & a & \text{RHS} \\ \hline z & 1 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ s & 0 & 1 & 3 & 1 & 1 & 0 & 0 & 12 \\ a & 0 & 2 & 1 & -1 & 0 & -1 & 1 & 7 \end{array}$$

$$\begin{array}{c|ccccccc|c} z & 1 & 2 & 1 & -1 & 0 & -1 & 0 & 7 \\ s & 0 & 1 & 3 & 1 & 1 & 0 & 0 & 12 \\ a & 0 & 2 & 1 & -1 & 0 & -1 & 1 & 7 \end{array}$$

$$\begin{array}{c|ccccccc|c} z & d_1 & d_2^l & d_3^l & s & e^l & \text{RHS} \\ \hline z & 1 & 0 & 0 & 0 & 0 & 3 \\ s & 0 & 0 & 5/2 & 3/2 & 1 & 1/2 & 17/2 \\ d_1 & 0 & 1 & 1/2 & -1/2 & 0 & -1/2 & 7/2 \end{array}$$

$$\Delta = \min \begin{cases} 4 - 0 = 4 & = 7/2 \\ (12 - 0)/1 = 12 \\ (7 - 0)/2 = 7/2 \end{cases}$$

$$\begin{bmatrix} s \\ a \end{bmatrix} = \begin{bmatrix} 12 \\ 7 \end{bmatrix} - \begin{bmatrix} 1 \\ 2 \end{bmatrix} \cdot \frac{7}{2} = \begin{bmatrix} 17/2 \\ 0 \end{bmatrix} \rightarrow \text{leave}(a)$$

$$z = 7 - 7/2 \cdot 2 = 3$$

$$d_1 = 0 + 7/2 = 7/2$$

$$d_2 = -2, d_3 = 2, s = 17/2, e = 0 \Rightarrow z$$

→ Phase 1 end.

Phase 2. Minimize $z = -2d_1 - 3d_2 + 2d_3$

$$\begin{array}{c|ccccccc|c} z & d_1 & d_2^l & d_3^l & s & e^l & \text{RHS} \\ \hline z & 1 & 2 & 3 & -2 & 0 & 0 & 3 \\ s & 0 & 0 & 5/2 & 3/2 & 1 & 1/2 & 17/2 \\ d_1 & 0 & 1 & 1/2 & -1/2 & 0 & -1/2 & 7/2 \end{array}$$

$$\begin{array}{c|ccccccc|c} z & 1 & 0 & 2 & -1 & 0 & 1 & 3 \\ s & 0 & 0 & 5/2 & 3/2 & 1 & 1/2 & 17/2 \\ d_1 & 0 & 1 & 1/2 & -1/2 & 0 & -1/2 & 7/2 \end{array}$$

$$\begin{array}{c|ccccccc|c} z & d_1 & d_2 & d_3^l & s^l & e^l & \text{RHS} \\ \hline z & 1 & 0 & 2 & -1 & 0 & 1 & -19/5 \\ d_2 & 0 & 0 & 5/2 & 3/2 & 1 & 1/2 & 7/5 \\ d_1 & 0 & 1 & 1/2 & -1/2 & 0 & -1/2 & 9/5 \end{array}$$

$$\begin{array}{c|ccccccc|c} z & 1 & 0 & 0 & -11/5 & -4/5 & 3/5 & -19/5 \\ d_2 & 0 & 0 & 1 & 3/5 & 2/5 & 1/5 & 7/5 \\ d_1 & 0 & 1 & 0 & -4/5 & -1/5 & -3/5 & 9/5 \end{array}$$

$$\Delta = \min \begin{cases} 3 - (-2) = 5 & = 17/5 \\ (17/2 - 0)/5/2 = 17/5 \\ (7/2 - 0)/1/2 = 7 \end{cases}$$

$$\begin{bmatrix} s \\ d_1 \end{bmatrix} = \begin{bmatrix} 17/2 \\ 7/2 \end{bmatrix} - \begin{bmatrix} 5/2 \\ 1/2 \end{bmatrix} \cdot \frac{17}{5} = \begin{bmatrix} 0 \\ 9/5 \end{bmatrix} \rightarrow \text{leave}(s)$$

$$z = 3 - 2 \cdot 17/5 = -19/5$$

$$d_2 = -2 + 17/5 = 7/5$$

$$\Delta = \min \begin{cases} \infty & = 11/3 \\ (7/5 - (-2))/1/5 = 17 \\ (9 - 9/5)/3/5 = 11/3 \end{cases}$$

$$\begin{bmatrix} d_2 \\ d_1 \end{bmatrix} = \begin{bmatrix} 7/5 \\ 9/5 \end{bmatrix} - \begin{bmatrix} 1/5 \\ -3/5 \end{bmatrix} \cdot \frac{11}{3} = \begin{bmatrix} 2/3 \\ 4 \end{bmatrix} \rightarrow \text{leave}(d_1)$$

$$z = -19/5 - 3/5 \cdot 11/3 = -6$$

$$e = 0 + 11/3 = 11/3$$

	z	x_1^u	x_2	x_3^l	s^l	e	RHS
z	1	0	0	-11/5	-4/5	3/5	-6
x_2	0	0	1	3/5	2/5	1/5	2/3
e	0	1	0	-4/5	-1/5	-3/5	11/3

z	1	1	0	-3	-1	0	-6
x_2	0	1/3	1	1/3	1/3	0	2/3
e	0	-5/3	0	4/3	1/3	1	11/3

$\rightarrow$ optimal ... $\begin{matrix} (min) \\ z^* = -6 \end{matrix}$ $\begin{matrix} (max) \\ \text{still} \end{matrix}$ $\underline{z^* = 6}$
 $x_2 = 2/3$
 $x_1(ub) = 4$
 $x_3(lb) = 2$

[5.25] Solve the following problem by the simplex method for bounded variables:

#8

$$\begin{aligned} \text{Minimize } & 2x_1 + 6x_2 - x_3 - 4x_4 + x_5 \\ \text{subject to } & 2x_1 + x_2 + 4x_3 + x_4 + x_5 = 10 \\ & 2x_1 + 8x_2 - 3x_3 + x_4 = 7 \\ & 0 \leq x_1 \leq 3 \\ & 1 \leq x_2 \leq 4 \\ & 0 \leq x_3 \leq 8 \\ & 1 \leq x_4 \leq 2 \\ & 0 \leq x_5 \leq 4. \end{aligned}$$

$$\begin{aligned} \text{Min } & 2d_1 + 6d_2 - d_3 - 4d_4 + d_5 \\ \text{s.t. } & 2d_1 + d_2 + 4d_3 + d_4 + d_5 + d_1 = 10 \\ & 2d_1 + 8d_2 - 3d_3 + d_4 + d_2 = 7 \\ & 0 \leq d_1 \leq 3, \quad 1 \leq d_2 \leq 4, \quad 0 \leq d_3 \leq 8 \\ & 1 \leq d_4 \leq 2, \quad 0 \leq d_5 \leq 4. \end{aligned}$$

$$\begin{aligned} X_{BV} &= \{d_1, d_2\} \text{ Initial BFS doesn't exist.} \\ d_1^l &= 0, \quad d_2^l = 1, \quad d_3 = 2, \quad d_4^l = 1, \quad d_5^l = 0. \\ d_1^l &= 0, \quad d_2 = 4 \\ \rightarrow X_{BV} &= \{d_3 = 2, d_2 = 4\} \\ \rightarrow & \text{a1 x12} \end{aligned}$$

Phase 1. Min $d_1 + d_2$

	z	d_1^l	d_2^l	d_3	d_4^l	d_5^l	d_2	RHS
z	1	0	0	0	0	0	-1	4
d_3	0	2	1	4	1	1	0	2
d_2	0	2	8	-3	1	0	1	4

$$\Delta = \begin{cases} 4-1=3 = \frac{16}{35} \\ \frac{2-0}{1/4} = 8 \\ \frac{4-0}{35/4} = \frac{16}{35} \end{cases}$$

	z	d_1^l	d_2^l	d_3	d_4^l	d_5^l	d_2	RHS
z	1	7/2	35/4	0	7/4	3/4	0	4
d_3	0	1/2	1/4	1	1/4	1/4	0	2
d_2	0	7/2	35/4	0	7/4	3/4	1	4

$$\begin{bmatrix} d_3 \\ d_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix} - \begin{bmatrix} 1/4 \\ 35/4 \end{bmatrix} \cdot \frac{16}{35} = \begin{bmatrix} 66/35 \\ 0 \end{bmatrix} \rightarrow \text{leave}(d_2) \text{ (L)} \text{ (BV)}$$

	z	d_1^l	d_2	d_3	d_4^l	d_5^l	RHS
z	1	0	0	0	0	0	0
d_3	0	2/5	0	1	1/5	16/65	66/35
d_2	0	2/5	1	0	1/5	3/35	51/35

$$\begin{aligned} d_2 &= 1 + \frac{16}{35} = \frac{51}{35} \\ z &= 4 - \frac{35}{35} \cdot \frac{16}{35} = 0 \end{aligned}$$

$$\begin{aligned} d_1 &= 0, \quad d_2 = 51/35, \quad d_3 = 66/35, \quad d_4 = 1, \quad d_5 = 0 \\ \text{(L)} \quad \quad \quad \text{(BV)} \quad \quad \quad \text{(BV)} \quad \quad \quad \text{(L)} \quad \quad \quad \text{(L)} \end{aligned}$$

$\rightarrow$ Phase 1 end

Phase 2. Min $2d_1 + 6d_2 - d_3 - 4d_4 + d_5$

	z	d_1^l	d_2	d_3	d_4^l	d_5^l	RHS
z	1	-2	-6	1	4	-1	20/7
d_3	0	2/5	0	1	1/5	16/65	66/35
d_2	0	7/5	1	0	1/5	3/35	51/35

$$\Delta = \begin{cases} 2-1=1 = 1 \\ (66/35-0)/1/5 = \frac{66}{7} \\ (51/35-1)/1/5 = \frac{16}{7} \end{cases}$$

	z	d_1^l	d_2	d_3	d_4^l	d_5^l	RHS
z	1	0	0	0	5	-33/65	20/7
d_3	0	2/5	0	1	1/5	16/65	66/35
d_2	0	7/5	1	0	1/5	3/35	51/35

$$\begin{bmatrix} d_3 \\ d_2 \end{bmatrix} = \begin{bmatrix} 66/35 \\ 51/35 \end{bmatrix} - \begin{bmatrix} 1/5 \\ 1/5 \end{bmatrix} = \begin{bmatrix} 59/35 \\ 44/35 \end{bmatrix} \rightarrow \text{no leave.}$$

$$\begin{aligned} d_4 &= 1+1=2 \rightarrow \text{ub.} \\ z &= 20/7 - 5 \cdot 1 = -15/7 \end{aligned}$$

	z	d_1^l	d_2	d_3	d_4^l	d_5^l	RHS
z	1	0	0	0	5	-33/65	-15/7
d_3	0	2/5	0	1	1/5	16/65	59/35
d_2	0	7/5	1	0	1/5	3/35	44/35

$\rightarrow$ optimal $z^* = -15/7$

$$d_2 = 44/35$$

$$d_3 = 59/35$$

$$d_1 = 0, \quad d_4 = 2, \quad d_5 = 0$$

Since $z_1 - C_1 = 0$, alternative optimal solution exists.

$$\text{For } z^* = -15/7, \quad d_1 = 9/14, \quad d_3 = 10/7, \quad d_2 = 1, \quad d_4 = 2, \quad d_5 = 0$$

is also an optimal solution

Make d_1 enter the basis

$$\Delta = \begin{cases} 3-0=3 = \frac{9}{14} \\ (59/35-0)/2/5 = \frac{59}{14} \\ (44/35-1)/2/5 = \frac{9}{14} \end{cases}$$

$$\begin{bmatrix} d_3 \\ d_2 \end{bmatrix} = \begin{bmatrix} 59/35 \\ 44/35 \end{bmatrix} - \begin{bmatrix} 2/5 \\ 2/5 \end{bmatrix} \cdot \frac{9}{14} = \begin{bmatrix} 10/7 \\ 1 \end{bmatrix} \rightarrow d_2 = 1 \rightarrow \text{lb.} \quad d_1 = 0 + 9/14 = 9/14$$